# Efficient and Controllable Remote Sensing Fake Sample Generation Based on Diffusion Model

Zhiqiang Yuan, Chongyang Hao, Ruixue Zhou, Jialiang Chen, Miao Yu, Wenkai Zhang, Hongqi Wang, and Xian Sun, *Senior Member, IEEE*

***Abstract*— In this article, we propose an efficient remote sensing fake sample generation (RSFSG) framework based on the diffusion model, so as to generate controllable samples consistent with real scenes. Firstly, in order to alleviate the huge time consumption caused by the large parameters of the diffusion model, we come up with a multifrequency dynamic knowledge distillation based on the consistent power spectrum of predicted Gaussian noise. The proposed multi-frequency knowledge transfer lets the lightweight model learn different frequency outputs from teacher model during diffusion process at different stages. Secondly, to address the problem of slow training of diffusion models, we propose a progressive training strategy (PTS), inspired by the fitting mechanism of deep networks from low to high frequencies. PTS enables fast fitting of diffusion models by enabling the model to learn from low-frequency information such as color at low resolution, and gradually move to high-resolution images full of details such as texture. The established two methods above achieve good generation performance under lightweight parameters, within almost half of the training time consumed. Extensive evaluations demonstrate that the proposed model significantly outperforms the state-of-the-art methods on RS controllable fake sample generation. Within our knowledge, we are the first to introduce the diffusion model into the RSFSG task and obtain good performance; the code and the corresponding pretrained files have been released at https://github.com/xiaoyuan1996/Controllable-Fake-Sample-Generation-for-RS.**

***Index Terms*— Diffusion model, multifrequency dynamic diffusion knowledge distillation, progressive training strategy (PTS) for accelerated diffusion learning, remote sensing fake sample generation (RSFSG).**

## I. Introduction

IN RECENT years, the research and application of artificial intelligence in the field of remote sensing has brought a huge impact on both military and civilian fields [1], [2], [3], [4]. However, the annotation performance of current data limits the acquisition of a large number of samples, which further restricts the performance of deep models [5], [6], [7]. Automatically, remote sensing fake sample generation (RSFSG) has gradually become a popular research direction that researchers are concerned about [8].

RSFSG is a method using for RS image batch inflation that can be further utilized for research in deep learning [9], [10], [11]. As a branch of image generation tasks, RSFSG is still in the early research stage. The early image generation tasks are based on deep generation models, which can be divided into two major categories according to the different representations of probability distributions. One is likelihood-based generative models [12], which learn the probability distribution of the data through the maximum likelihood criterion, such as autoregressive models [13], streaming models [14], and variational auto-encoder (VAE) [15]. The second is implicit generative models [16], in which the probability distribution is not expressed explicitly, such as generative adversarial networks (GANs) [17], which enable generators to achieve a direct transformation from noise to samples by adversarial learning.

In the past two years, the diffusion model [18] has gradually shown its advantages, which has a more flexible architecture

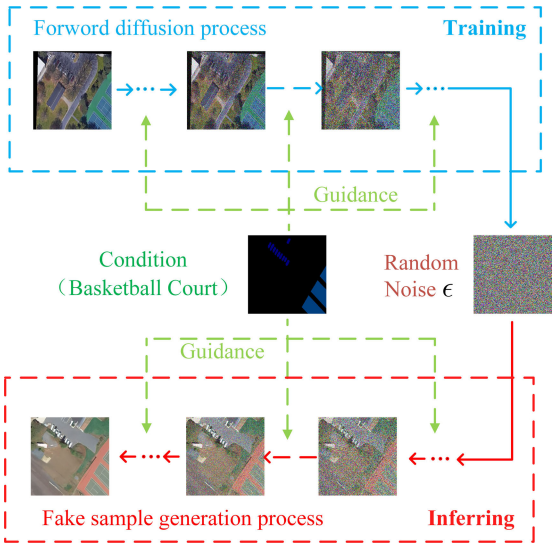



Fig. 1. Framework of RSFSG based on the diffusion model. The blue line part of the diagram represents the forward diffusion process, while the red line part of the diagram represents the pseudo-sample generation phase.

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and accurate log likelihood calculation compared with GAN and VAE. Diffusion models learn information decay due to noise, and then use the learned patterns to generate images, which has excellent performance in many application areas, such as computer vision [19], multimodal modeling, etc. [20], and even outperforms GAN and VAE in the field of sample generation. The emergence of the diffusion model provides an opportunity and possibility for good RSFSG like the framework of Fig. 1.

Although diffusion models have achieved great success in the past few years, due to the large number of parameters, they take a long time to train and infer when generating high-resolution RS images. In order to alleviate this problem, this article starts with the lightweight model and the training strategy to optimize the conventional diffusion model. On the one hand, although lightweight parameters can achieve faster operating efficiency, due to the shortcomings of its fitting ability, it cannot obtain a more accurate noise estimation like the huge parameter model. Moreover, we found that in the case of a small number of parameters, the noise output by the diffusion model obviously has a greater difference at the different sampling period. To improve the noise estimation ability of the lightweight model, we attempt to make the model adaptively learn different frequency signals from the teacher network, so as to obtain a noise distribution with consistent output power. On the other hand, the conventional diffusion model tends to use a fixed resolution for model training, while the non-dynamic resolution results in extremely long training time consumption when training a large number of iterations. Inspired by the deep network from low- to high-frequency fitting, we consider about whether the diffusion model can start from low-frequency information such as color and gradually learn high-frequency features such as texture. For this purpose, we design a progressive training strategy (PTS) to achieve controllable training from easy to hard by learning from low to high resolution. The established two methods above achieve a good generation performance under lightweight parameters, within only half of the training time consumed.

In general, the main contributions of our research are as follows.

1) We propose a controllable and efficient framework for an RSFSG-based diffusion model. Within our knowledge, we are the first to introduce the diffusion model into the RSFSG task and obtain superior performance.
2) We come up with a multifrequency dynamic knowledge distillation based on the consistent output power of the diffusion model. The proposed transfer method achieves competitive results by balancing the frequency band output of lightweight models.
3) To address the problem of slow training of diffusion models, we propose a PTS to speed up the diffusion process by going from low-frequency color to high-frequency texture.

Extensive evaluations demonstrate that the proposed model significantly outperforms the state-of-the-art methods on RS controllable fake sample generation. The rest of the article is organized as follows: Section II briefly summarizes the relevant work related to this study. Next, in Section III, we detail the efficient diffusion model learning method and training strategy. In Section IV, we conduct a large number of experiments to provide helpful qualitative and quantitative data for RSFSG task. Finally, the conclusion is given in Section V.

## II. Related Works

This section first introduces the development and related achievements of the image generation task in the field of natural scenes and remote sensing, and then summarizes the progress of the diffusion model.

### A. Evolution of Image Generation

We will introduce the evolution of image generation from natural scenes and remote sensing.

*1) Image Generation in Natural Scenes:* The rapid development of the field of image generation is inseparable from the contributions of the two classic models of GAN [17] and VAE [15]. In 2014, GAN restored random noise to clear images through generative adversarial training, which made a huge breakthrough in the field of image generation. After that, CGAN [21] added additional conditional constraints to the generator and discriminator of the GAN model so that the model can generate class-specific images. In 2015, Radford et al. [22] introduced convolutional neural networks into GANs and achieved more dramatic results in sample generation. Sohn et al. [23] proposed the CVAE model, which modifies the function of KL scatter to control the class of the generated images by controlling the mean, and obtained more promising results on the MNIST [24] dataset. VGAE [25] applied the VAE model to the field of image generation, which uses the two-layer GCN [26] as an encoder and the decoder reconstructs the graph by calculating the probability of the edge between two points. Di and Patel [27] combined CVAE [23] and GAN to offer competitive results in face generation. VQ-VAE [28] incorporated the idea of vector quantization based on the VAE framework, which exploits discrete data to represent the encoder network output and then generates images by point-by-point regression. In the past two years, many excellent models based on deep generation models have achieved great success in natural scene image generation; however, there is a still deficiency in the research of RS image generation.

*2) Image Generation in Remote Sensing:* The image generation in RS is more focused on image fusion [29] to obtain high-definition multispectral RS images. Since different RS data have different spatial resolution, spectral resolution, and temporal phase resolution, it is challenging to obtain a high-quality RS dataset. In 2015, Huang et al. [30] experimented with DNN for panchromatic sharpening and used a stacked sparse denoising autoencoder algorithm to generate high-resolution multispectral images. Palsson et al. [31] generated hyperspectral high-resolution images using a 3-D convolutional neural network, which firstly reduced the dimension of the hyperspectral images, and then fused the multispectral images for sample generation. The multiscale and multidepth

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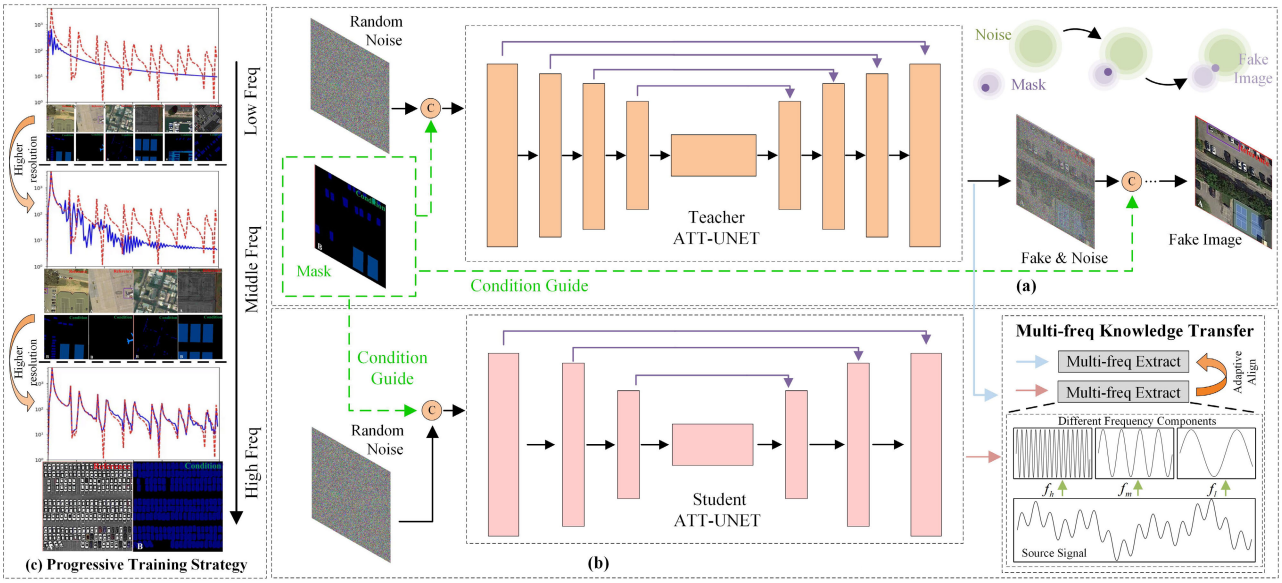



Fig. 2. Proposed framework for efficient and controllable RSFSG. (a) Original reverse diffusion process. (b) Designed diffusion distillation based on multifrequency knowledge transfer. (c) PTS for accelerated diffusion learning.

CNN proposed by Yuan et al. [32] focuses on improving the spatial resolution of multispectral images. The DHSIS [33] approach is mainly dedicated to fuse multispectral images with different spatial resolutions in the same scene to obtain HR-HSI. In 2020, Liu et al. [34] invented the PSGAN network to solve the problem of general sharpening of RS images. The GAN-RSIGM algorithm [35] has contributed greatly to the creation of high-resolution RS image samples by using the Wasserstein distance to reduce the gap between the generated samples and the real image. Compared to the above research methods, the RS image pseudo-sample generation method studied in this article, where the input data are a category mask, is more controllable and the generated sample set can be used for multidirectional research.

### B. Development of Diffusion Model

Diffusion model [36] is a recently brilliant generation model, which has achieved great successful in research directions such as graphical conversion and natural image generation. Compared with the GAN, the diffusion model eliminates the discriminator and the loss function is mathematically derived as a standard convex cross-entropy loss function, which assists the model training process simpler. The diffusion model is divided into the process of noise addition and denoising, and the purpose is to optimize the relevant parameters in the denoising process so as to make the generated images more realistic. Sohl-Dickstein et al. [37] presented the backward diffusion process and the prototype of the data generation model, while the article also lays the foundation for the mathematical theory related to the diffusion model. Denoising diffusion probabilistic models (DDPMs) [36] introduced the denoising score matching function into the probabilistic model, and obtained a good performance in the unconditional CIFAR10 [38] dataset. In the same year, Song et al. [39] presented the denoising diffusion implicit models for the inference, which greatly improved the sample noise reduction speed while ensuring the quality of the generated images. After that, Nichol and Dhariwal [40] proposed the improved DDPM in 2021, which mainly changed the sample method and adopted an important sampling scheme for gradient noise. In 2022, Singh et al. [41] ameliorated the diffusion model structure by adding conditional constraint information to enable the model to output class-specific images. Although diffusion has received wide acclaim in the fields of graphic conversion and sample generation in natural scenes, it is rarely used in RS images.

## III. Method

This section first reviews the conditional diffusion model, then as shown in Fig. 2, the proposed multifrequency dynamic diffusion knowledge distillation and PTS for accelerated diffusion learning are introduced in detail.

### A. Conditional Diffusion Model

The conditional diffusion model is on the basis of the DDPM. For a dataset $\mathcal{D} = \{x_i\}_{i=0}^{N}$, DDPM learns the data distribution $p(x)$ and generates samples consistent with the distribution. Different from DDPM, given the dataset of data pairs, denoted $\mathcal{D} = \{x_i, y_i\}_{i=0}^{N}$, conditional DDPM learns the distribution $p(y|x)$. During inference, conditional DDPM generates a sample according to the given $y$. The characteristic of conditional DDPM indicates its potential in many downstream tasks, including super-resolution, deblurring, and dehazing.

The essential idea of DDPM and conditional DDPM is to learn a network to generate samples from random Gaussian noise by denoising iteratively. Before that, we first go through a Gaussian diffusion process, where the input image $x_0 \sim q(x)$ is added Gaussian noise gradually in $T$ steps until the noisy image $x_T$ becomes an isotropic Gaussian distribution, namely $x_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. The process is a Markov chain and given $x_{t-1}$, the probability of $x_t$ is a Gaussian distribution

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with mean $(1-\alpha_t)\mathbf{I}$ and variance $\sqrt{\alpha_t}x_{t-1}$, described as

$$q(x_t|x_{t-1}) := \mathcal{N}(x_t; \sqrt{\alpha_t}x_{t-1}, (1-\alpha_t)\mathbf{I}) \quad (1)$$

$$q(x_t|x_0) = \prod_{t=1}^{t} q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{\gamma_t}x_0, (1-\gamma_t)\mathbf{I}) \quad (2)$$

where the $\{\alpha_t \in (0,1)\}_{t=1}^{T}$ linearly increases as $t$ increases, controlling the added Gaussian noise in every step, and $\gamma_t = \prod_{i=1}^{t}\alpha_i$. When $T \to \infty$, $x_T$ becomes the standard Gaussian noise. To make the mean and variance differentiable, given (1), we have

$$\begin{aligned} x_t &= \sqrt{\alpha_t}x_{t-1} + \sqrt{1-\alpha_t}\epsilon \\ &= \sqrt{\gamma_t}x_0 + \sqrt{1-\gamma_t}\bar{\epsilon}_t \end{aligned} \quad (3)$$

where $\epsilon, \bar{\epsilon}_t \sim \mathcal{N}(0, \mathbf{I})$, and $\bar{\epsilon}_t$ represents the corresponding value sampled from standard Gaussian noise for generating $x_t$.

If we reverse the above process, then the Gaussian noise $\hat{x}_T \sim \mathcal{N}(0, \mathbf{I})$ will be removed iteratively to obtain an image $\hat{x}_0$, which is referred to as *sampling*. For conditional DDPM, in addition to the Gaussian noise input, we also provide the conditions to constrain the diversity in denoising. To learn the reverse process $q(\hat{x}_{t-1}|\hat{x}_t)$, we let the model $p_\theta$ to approximate the probability

$$p_\theta(\hat{x}_{t-1}|\hat{x}_t, c) := \mathcal{N}(\mu_\theta(\hat{x}_t, c, \gamma_t), \Sigma_\theta(\hat{x}_t, c, \gamma_t)) \quad (4)$$

where $c$ represents the condition of guiding the reverse diffusion. Although $q(x_{t-1}|x_t)$ is hard to estimate, $q(x_{t-1}|x_t, x_0)$ can be obtained with $q(x_t|x_0)$ and $q(x_t|x_{t-1})$. Also, there exists a variational lower bound for $\log p_\theta(x_0)$, allowing us to optimize the $p_\theta(x_{t-1}|x_t)$ with $q(x_{t-1}|x_t, x_0)$. The reverse process can be learned with a model $f_\theta$. Given a noisy image $\hat{x}_t$, we assume $\hat{x}_t$ is an intermediate image of the forward diffusion

$$\hat{x}_t = \sqrt{\gamma_t}\hat{x}_0 + \sqrt{1-\gamma_t}\bar{\epsilon}_t. \quad (5)$$

The model $f_\theta$ aims to estimate the added noise $\bar{\epsilon}_t$ in the forward diffusion so as to recover the $\hat{x}_0$.

In our conditional DDPM, we take the segmentation mask as the condition $c$, the model $f_\theta(\hat{x}_t, c, \gamma)$ takes the noisy image $\hat{x}_t$ and the condition $c$ as the input, with the hyperparameter $\gamma_t$ in forward diffusion. The objective for training $f_\theta$ aims to estimate the noise vector

$$E_{(x,y)}E_{\epsilon,\gamma}\|f_\theta(\sqrt{\gamma}x_0 + \sqrt{1-\gamma}\bar{\epsilon}, c, \gamma) - \epsilon\|_1^1. \quad (6)$$

During inference, given noise input $\hat{x}_t$, the recovered image $\hat{x}_0$ can be derived from (5)

$$\hat{x}_0 = \frac{1}{\sqrt{\gamma_t}}(\hat{x}_t - \sqrt{1-\gamma_t}f_\theta(\hat{x}_t, c, \gamma_t)). \quad (7)$$

Recall that we can represent $q(x_{t-1}|x_t, x_0)$ with $q(x_t|x_0)$ and $q(x_t|x_{t-1})$, with Bayes' rule, we can parameterize the mean of the reverse diffusion as

$$\mu_\theta(\hat{x}_t, c, \gamma_t) = \frac{1}{\sqrt{\alpha_t}}(\hat{x}_t - \frac{1-\alpha_t}{\sqrt{1-\gamma_t}}f_\theta(\hat{x}_t, c, \gamma_t)). \quad (8)$$

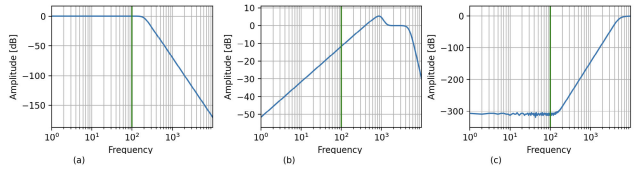



Fig. 3. Frequency responses of various filters in the proposed multifrequency knowledge transfer. Hand-extracted multiband features allow the model to adaptively complement the missing bands to obtain a more uniform noise estimate. (a) Low-frequency filter response. (b) Mid-frequency filter response. (c) High-frequency filter response.

Following this equation, if we use $\Sigma_\theta(\hat{x}_t, c, \gamma_t) = \sqrt{1-\alpha_t}$, the update of $\hat{x}_{t-1}$ can be modeled as:

$$\hat{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}}(\hat{x}_t - \frac{1-\alpha_t}{\sqrt{1-\gamma_t}}f_\theta(\hat{x}_t, c, \gamma_t)) + \sqrt{1-\alpha_t}\bar{\epsilon}_t \quad (9)$$

where $\bar{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.

### B. Multifrequency Dynamic Diffusion Knowledge Distillation

Although the diffusion model has achieved good results in generating tasks, it takes a long time to train and infer due to the large number of parameters. Automatically, designing a student model with lightweight parameters and maintaining the performance of the original model has become an urgent area for optimization. For lightweight models, the reduction in parameters makes it difficult to obtain power spectral densities consistent with the standard Gaussian due to the poor fit capability. The noise estimates learned by the diffusion model with a large number of parameters are anisotropically uniform [see Fig. 8(a)]; however, we found that in the case of a small number of parameters, the noise output by the lightweight diffusion model obviously has unbalanced frequencies at the late sampling period [see Fig. 8(b)]. For this purpose, we come up with a multifrequency dynamic knowledge distillation method, which improves the noise prediction of the student network by dynamically aligning the power density to the teacher network in different frequency bands.

Firstly, for the noises $\hat{\epsilon}^t$ and $\hat{\epsilon}^s$ predicted by the teacher network and the student network, we design a multifrequency extract module to obtain their frequency features at different levels. Specifically, for the predict noise $\hat{\epsilon}$, we first denote different-of-gaussian [52] as $\mathbb{D}$, where $\mathbb{D}$ is the difference of two gaussian filters:

$$\mathbb{D}(k_1, k_2) = \mathcal{G}_{k_1\times k_1}(\epsilon) - \mathcal{G}_{k_2\times k_2}(\epsilon) \quad (10)$$

where $\mathcal{G}_{k\times k}(\cdot)$ denotes the Gaussian filter with kernel size of $k$. $\mathbb{D}$ represent the different frequency components, and can be used to produce the high, middle and low frequency features $f_h$, $f_m$ and $f_l$ with different $k$. $f_h$, $f_m$, $f_l$ can be regarded as high-pass, mid-pass and low-pass filtered signals, which can be used as components to apply aligning.

Furthermore, in order to achieve dynamic frequency alignment between the two networks, we set three trainable weights $\beta_l, \beta_m, \beta_h$ to achieve adaptive loss calculation

$$\mathcal{L}_{mkt} = \beta_l\mathcal{L}_1(f_l^s, f_l^t) + \beta_m\mathcal{L}_1(f_m^s, f_m^t) + \beta_h\mathcal{L}_1(f_h^s, f_h^t) \quad (11)$$

where $\mathcal{L}_1(x, y)$ denotes the $L_1$ loss of features $x$ and $y$.

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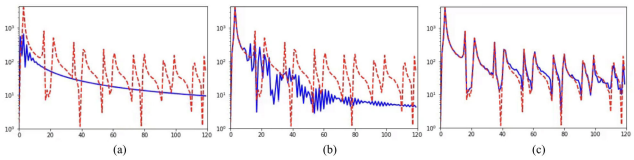



Fig. 4. Red line is the target distribution, and the blue line is the model prediction. The evolution of deep network representation has a clear law in the frequency domain space: the frequency converges sequentially (a)–(c) from low to high.

We show the frequency response of the designed different frequency filters $f_l$, $f_m$, $f_h$ in Fig. 3, respectively. $f_l$, $f_m$, $f_h$ manually extract noises in the low-frequency, mid-frequency, and high-frequency bands, respectively, to provide relevant components for dynamic learning of lightweight networks. The trainable parameter $\beta$ gives the student network the ability to adaptively adjust the missing frequency band, so that the lightweight network can supplement the output through the extracted different frequency information. By enabling the student network to adaptively learn the frequency components in the teacher network, the multifrequency dynamic diffusion knowledge distillation method allows the diffusion model to obtain a more uniform power spectrum to improve noise prediction accuracy.

### C. PTS for Accelerated Diffusion Learning

Although the diffusion model achieves good performance on tasks such as image generation, it is undoubtedly time consuming as DDPM needs to be trained at multiple steps. As shown in Fig. 4, in traditional deep networks, the model tends to start learning at low frequencies, progressively learning the components of functions at higher frequencies [42]. Inspired by this, we conjecture that this law also exists in the diffusion model, that is, first learn the low-frequency information of the generated image such as local color, and gradually learn the high-frequency information such as texture. If such a law exists, then we can control the learning efficiency of the model at different frequencies, the easy-to-learn areas such as low-frequency areas can be accelerated, while the difficult areas such as high frequencies can be learned more intensively.

In order to verify our conjecture, we set up a PTS to make the model focus on the difficult areas in the learning process. Specifically, for the sample $\mathscr{S}$, we resize it to a smaller size $\mathscr{S}_1$ to reduce the details in the sample in the early stage of learning, forcing the model to learn low-frequency information such as hue, object distribution, etc. After time $\mathscr{T}_1$, we resize sample $\mathscr{S}$ to a larger size than $\mathscr{S}_1$ to enable the model to learn more fine-grained frequency features. At the final $\mathscr{T}_N$ moment, we release the restrictions on model learning so that the model starts to learn high-frequency information such as texture in the sample. Through the adjustment of $\mathscr{T}_n$, the learning efficiency adjustment at different stages is realized, thus allowing the rapid convergence of the diffusion model.

The designed PTS starts from the property that neural networks preferentially use low frequencies to fit the objective function and progressively learn the different frequency components of the sample, thus converging to high frequencies. Through different treatments of areas with inconsistent difficulty, the learning degree of the model at different frequencies is adjusted, so as to achieve more efficient training efficiency. However, PTS allows the diffusion model to cover more scale training samples, so that the model has better reconstruct performance for samples with multiscale characteristics in remote sensing. Section IV-F in the experiment greatly verifies the effectiveness of PTS in improving training time.

## IV. Experiment and Analysis

In this section, we first describe the dataset, configuration, and implementation details, and then qualitative and quantitative experiments are carried out to illustrate the effectiveness of the proposed method.

### A. Datasets and Indicators

Different from scene segmentation, the task of fake sample generation needs to judge the recovery performance from the situation of multiscale, zero information, etc. To provide a general baseline for controllable RSFSG, we select more representative and scale-specific samples for training and testing based on the iSAID dataset [43], and ensure the integrity of large targets such as sports fields and the resolution of small targets. We select eight types of targets, such as Plane, Small Vehicle, and Ground Track Field, with 1411 high-resolution RS images as the initial training data to achieve rapid training and evaluation. Furthermore, 453 RS images with the same specifications and corresponding mask are included in the test set, involving eight target types with different mask information, and more than two hundred images to achieve a multilabel diversified target scenes. In addition to providing a mutilated target mask for model edge calculation capability, 46 pure background plate images are also included to evaluate the scene recovery capability of the model.

In this experiment, we use four indicators to evaluate the model performance.

1) *IS* ↑ *[44]:* The IS indicator evaluates the image quality by calculating the definition and diversity of the generated image and uses the entropy of the edge distribution of the generated image in all categories to describe sample diversity. Finally, the image quality is calculated by calculating their KL scatter, and the larger the distance of their probability distribution, the better the image quality.
2) *FID* ↓ *[45]:* Frechet Inception Distance score is often used as one of the evaluation indicators of the generation model to evaluate the quality of generated samples. FID is a method to measure the similarity of two sets of image datasets, and a measure to calculate the distance between the feature vectors of the real image and the generated image.
3) *MUSIQ* ↑ *[46]:* A multiscale image quality transformer method based on image blocks, which processes full-size

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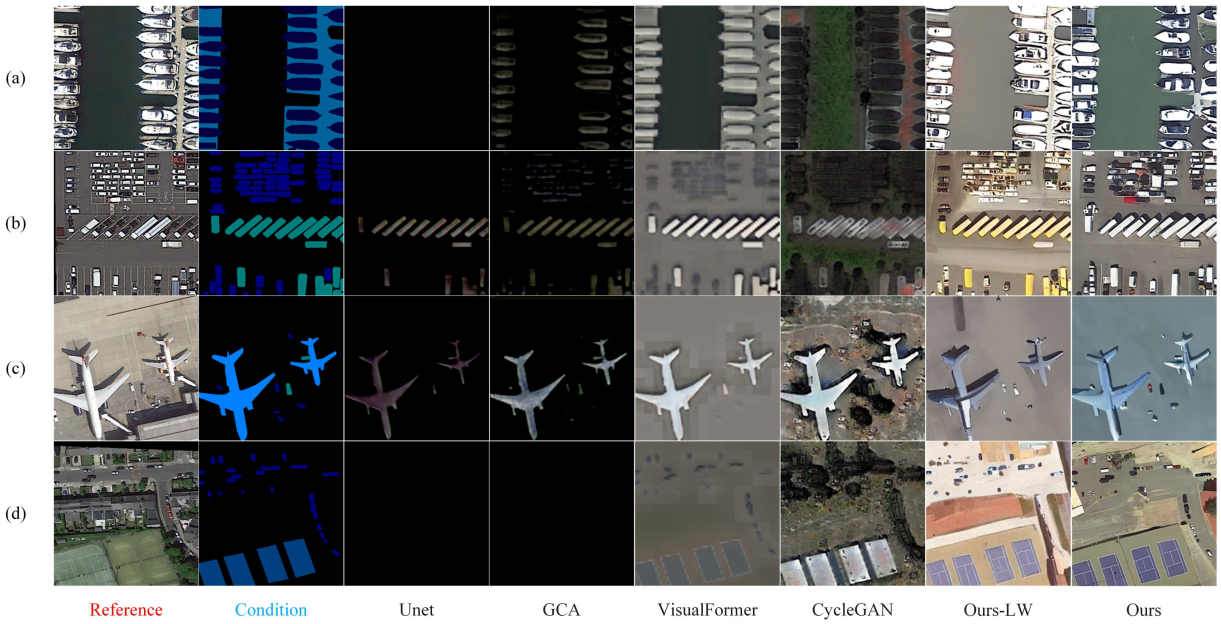



Fig. 5. Qualitative comparison of different models. We select four examples of images, covering five target types: plane, ship, small vehicle, large vehicle, courts. (a)-(d) are four examples.

input images of different sizes and different aspect ratios. The method evaluates image quality by simultaneously extracting multiscale features while maintaining aspect ratio scaling.

4) *NIMA* ↑ *[47]:* NIMA algorithm generates a rating histogram for any image and directly compares images of the same subject.

The four indicators evaluate the performance from the aspect of reference and no-reference, so as to have a fairer evaluation.

### B. Implement Details

All experiments in this article are conducted on eight NVIDIA RTX 3090 GPUs. We first uniformly crop large images to 256 × 256 resolution as network inputs and use data enhancement methods, such as flipping and mirroring, to improve the robustness of the model. The diffusion model is initialized with the linear parameter method, the step is 2000, and the value interval is $1e^{-6}$ to $1e^{-2}$. The backbone network uses att-UNet, which contains a 16-layer resnet network, an eight-layer residual network with an attention mechanism, and a five-layer downsampling network and a five-layer upsampling network. For the PTS, we divide the training into four stages, and we allocate more time in the later stage for clearer generation details. In the early stage, it is divided into two stages, and the input resolution is set to 64 × 64 and 128 × 128, respectively. The training times in both stages account for 20%, mainly learning low-frequency information such as hue and object distribution. In the mid stage, the resolution is set to 196 × 196, and the training accounted for 20%, mainly learning features such as target outlines and color descriptions. In the later stage, we adjust the resolution to 256 × 256, and the training accounted for 40%, mainly learning high-frequency information such as target texture and background details. We employ the Adam optimizer with $L1$ loss to train the network with 150 W iterations. The batch size is adjusted to 4 and dropout rate is set to 0.2. To make the experimental evaluation more stable, we average the final five model parameters so as to get the fair performance.

### C. Comparisons With State-of-the-Art Methods

To verify the effectiveness of the proposed method on the task of controllable RSFSG, we conduct qualitative and quantitative comparisons with other leading models. The comparison model is as follows.

1) *Unet [48]:* Unet uses a fully convolutional neural network to stitch the number of channels to form thicker features, and is one of the most commonly used segmentation models.
2) *GCA [49]:* GCA network follows the basic automatic encoder structure. The model uses the latest smooth expansion technology to obtain the global prediction.
3) *VisualFormer [50]:* The VisualFormer uses Transformer for end-to-end image transformation, which is one of the most popular architectures in current research.
4) *CycleGAN [51]:* An improved version based on GANs, using cycle consistency loss to preserve the authenticity of generated samples.

For the four methods described above, we migrate the open source code to the RSFSG task, fine tune specific parameters, and perform-cross validation to obtain reliable results. The qualitative and quantitative results are shown in Fig. 5 and Table I, respectively.

Based on the analysis of qualitative and quantitative results, we conclude that the result performed by Unet is the worst. The inferred images are ambiguous, the target is specious, and the background is monotonous in black and white. The FID indicator is as high as 443.81, which is far from other methods. GCA model has little progress compared with Unet network, and the recognition of small targets such as ships has

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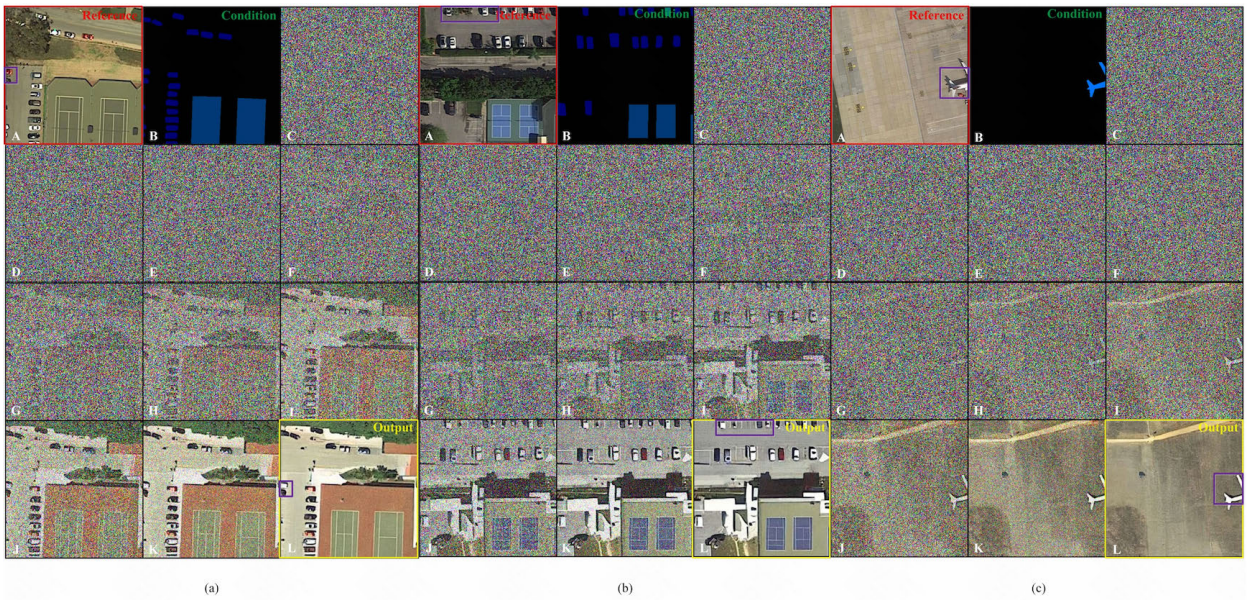



Fig. 6. Visualization of generated results. (a) is the GT sample, (b) is the mask, (c)-(k) are the intermediate results, (l) are the generated image.

TABLE I
QUANTITATIVE COMPARISON OF RESULTS GENERATED BY VARIOUS METHODS

| Method | FID ↓ | IS ↑ | MUSIQ ↑ | NIMA ↑ |
|---|---|---|---|---|
| Unet | 443.812 | 2.050 | 23.173 | 3.819 |
| GCA | 431.947 | 2.186 | 23.353 | 4.154 |
| VisualFormer | 300.300 | 3.900 | 26.369 | 3.955 |
| CycleGAN | 294.512 | 4.570 | 35.575 | 3.897 |
| Ours | **211.410** | **7.196** | **40.670** | **4.745** |
| Ours-LW | 220.858 | 6.528 | 32.720 | 4.744 |

TABLE II
QUANTITATIVE COMPARISON OF PARAMETERS AND TIME CONSUMPTION

| | Parameters (M) | Infer time (ms) |
|---|---|---|
| Ours | 93.276 | 70.707 |
| Ours-LW | 40.106 | 27.890 |

been improved, but the IS indicator has not been improved, and the image is still only in black and white. Compared with the first two models, the learning ability of VisualFormer has made significant progress, which has obvious generation traces with low resolution and small targets. However, there are still some problems such as monotonous image color, poor noise processing ability, and insufficient target details generation. CycleGAN is better at rendering the background and color of the image, the description of target details is still very rough. Our model achieves SoTA in both experimental metrics and visualization results, with the lowest FID value and IS value close to 7.2. In addition, the lightweight model with the proposed KD method also achieved the best performance with other deep networks. Through the analysis of the display results in Fig. 5, the visual effect is more consistent with the expected effect, and the clarity, target integrity, background, and color diversity of the output image are more consistent with the real RS images.




Fig. 7. Qualitative comparison under different lightweight training methods. (a) LW-d. (b) LW-a. (c) LW-mdd.

In addition, we quantitatively analyze the time consumption and parameter amount of the lightweight model as shown in Table II. Compared with the 93M parameters of the original DDPM, the lightweight model has only 40M parameters and reduces the consumption by about one-third in the inferring time. Also, in Section IV-F, we have reduced the training time consumption by about 50% through the PTS strategy, which greatly improved the training and inference efficiency of the model.

### *D. Visual Analysis of Inference Process*

Fig. 6 shows the three samples, which are divided into 12 portions. Image A represents the GT RS image, image B is the input mask, the input initialized random noise is in image C, and image L shows the output results. C to L reflect the denoising process of the diffusion model, we set the total number of lists to 10 and automatically adjust the distance to save the denoised image. The method of this study is realistic for multitarget reduction in the field of RS, and the background comparison between the reference images demonstrates that the diversity and clarity of the output images of this method reach a higher level. In addition, the part marked by the purple box in each sample proves that the present method has a

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TABLE III
QUANTITATIVE COMPARISON OF LIGHTWEIGHT MODEL WITH DIFFERENT CONFIGURATIONS

| Method | FID ↓ | IS ↑ | MUSIQ ↑ | NIMA ↑ |
|---|---|---|---|---|
| LW-d | 250.437 | 6.830 | 29.098 | 4.279 |
| LW-a | 245.000 | **7.044** | 31.797 | 4.651 |
| LW-mdd | **220.858** | 6.528 | **32.720** | **4.744** |

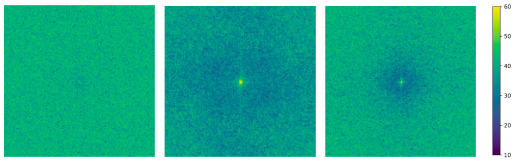



Fig. 8. Comparison of power spectrum before and after MDD distillation at the late diffusion process. The power spectrum of the teacher network is close to the normal distribution, while the distribution of the lightweight network trained with the MDD method is more uniform than without MDD. (a) Teacher. (b) Student (w/o MDD). (c) Student (with MDD).

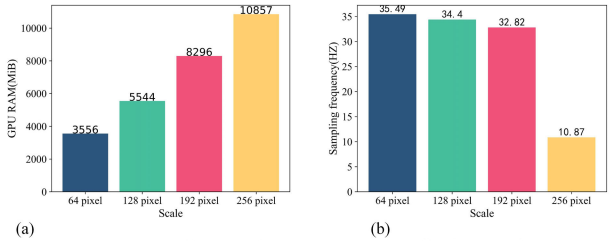



Fig. 9. Comparison of resource usage within different input sizes. (a) GPU resource consumption comparison. (b) Sampling rate comparison.

well-developed degree of recovery for the edge incomplete targets. Last but not least, not only the last step of the inverse diffusion can be used as the training sample in this research method, but also the result of the intermediate process with some noise attached can be taken as the negative sample. More visualization examples can be found in Fig. 12 of Appendix.

### *E. Ablation Analysis of Multifrequency Dynamic Distillation*

In this section, we verify the effectiveness of the proposed multifrequency dynamic distillation method for knowledge transfer of diffusion models. We regarded the directly trained lightweight network as the control group LW-d, the direct dynamic alignment of teacher and student output is used as the control group LW-a, and the lightweight model with our distillation method is denoted as LW-mdd.

The quantitative comparison of lightweight model with different configurations is shown in Table III. Compared with direct training and transfer methods, our distillation method achieves best performance on most quantization results, which is due to the ability of the MDD to equalize the power distribution in the student network thus allowing for a more uniform power output. We show the qualitative comparison under different lightweight training methods in Fig. 7. Compared with other control groups, our method produces samples with less noise and sharper boundaries.

We visualize the predicted noise power spectrum before and after adding MDD distillation method during training in Fig. 8(a). As shown in Fig. 8, in the later stage of the diffusion process, the output power of the teacher network is relatively uniform, which has a relatively consistent density in each frequency band. However, the direct training of the student network due to the lack of parameters leads to a decrease in fitting ability, making the corresponding spectral distribution in Fig. 8(b) concentrated at low frequencies and lack power distribution at high frequencies. Thanks to the ability of MDD of dynamically adjusting the consistency of frequency components, the power spectral density in Fig. 8(c) generated after adding the MMD distillation method is more uniform and more similar to the output of the teacher network.

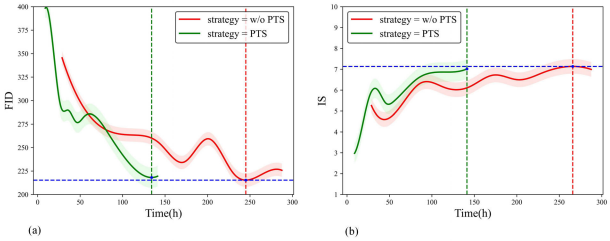



Fig. 10. Green curve represents the use of PTS strategy, the red curve represents the use of conventional training strategy. The dotted lines are the training time to reach the extreme value. Variation of (a) FID and (b) IS under different training strategies.

### *F. Performance Analysis of Progressive Training*

The slow sampling process and excessive resource consumption of diffusion model have actually been disturbing problems. The forward diffusion process often takes a lot of time to train the parameters, and even longer for high-resolution images as shown in Fig. 9. According to the PTS strategy, if we use small resolution in the early stage of training and use full resolution in the later stage of training, in this case we will be able to obtain the maximum training benefit with less training time. For this purpose, we conduct a comparative experiment of the PTS strategy to prove the availability of the proposed method in terms of operating efficiency.

The implementation of PTS is to learn low-frequency information first, and then gradually progress to feature extraction of high-frequency information. The size of the input tensor is increased following the learning progress, and the learning rate is adjusted for different scales by adjusting the appropriate learning rate and setting the appropriate training time. As shown in Fig. 10, we count the experimental indicators at different stages, and find that the training time is shortened by nearly half after using this strategy. According to the blue dashed line, the performance of both models is almost the same, which proves that the PTS strategy is not only effective in saving time but also in ensuring the generation performance.

The effectiveness of the PTS strategy implies that when training the diffusion model, it is possible to first learn low-resolution images to fit low-frequency information, and then continue to strengthen the results at the spatial global minimum. By increasing the resolution step by step, more multiscale training effects and lesser training resource consumption can be obtained.

### *G. Ablation Study on DDIM Sampling*

There are 2000 steps in the DDPM experiments, which causes us about 8 h to run a set of validation sets. Therefore,

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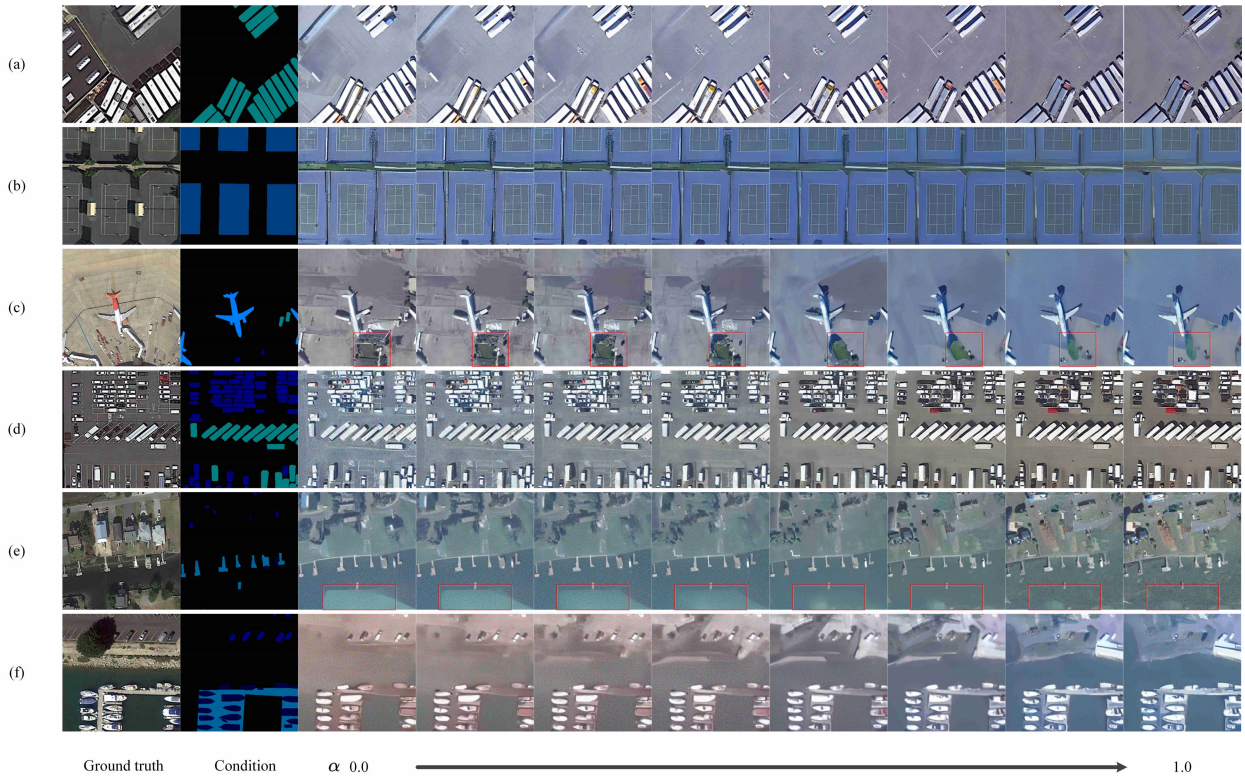



Fig. 11. Impact of initial noise on the generated results. (a) Original image. (b) Model input. (c)–(f) Output results when $\alpha$ step from 0 to 1.

TABLE IV
COMPARISON OF GENERATION EFFECTS UNDER DIFFERENT $T$ AND $\eta$

| $\eta$ | - | $T$ 5 | 10 | 20 | 50 | 100 |
|---|---|---|---|---|---|---|
| 0 | ↓ FID | 243.93 | 224.25 | **204.17** | 207.03 | 205.32 |
| | ↑ IS | 6.230 | 6.298 | 6.279 | 6.196 | **7.136** |
| 0.5 | ↓ FID | 249.12 | 235.09 | 221.91 | 212.78 | 214.62 |
| | ↑ IS | 5.908 | 5.978 | 6.284 | 5.965 | 6.033 |
| 1 | ↓ FID | 271.98 | 261.61 | 251.23 | 240.98 | 227.03 |
| | ↑ IS | 5.123 | 5.960 | 6.073 | 6.616 | 5.637 |

we adopt the continuous sampling of DDIM to accomplish high restoration of images by a small number of iterations. In DDIM [39], $\eta$ controls the rate from DDPM to DDIM, while step $T$ controls the sample quality. To study the parameter steps $T$ and $\eta$ on the final performance, we carry out the experiments as shown in Table IV.

Table IV shows that when $T$ is constant, the results become worse with the increase in $\eta$. The results are obviously improved with the increase of step $T$ when $\eta$ value is 0. However, after the step configuration exceeds 20, the FID indicator increases slightly. We observe that the number of restoration iterations suitable for the RS sample generation task is 20. If the number is lower than this value, the image generated will still have noise. If the number is higher than 20, the diversity of the image generated will be improved, but a large amount of time will be wasted. To sum up, setting $T$ and $\eta$ to 20 and 0.0, respectively, brings the highest cost performance, which can not only obtain more real RS images but also effectively shorten the time. The adjusted method only takes 30 min to reason in validation sets.

### H. Effect of Initial Noise Sensitivity

The initialization noise is also a critical factor affecting the quality of sample generation. To examine the effect of initialization noise on the results, we set a spherical function for the initialization noise and control the orthogonal scale of two random noises by a variable $\alpha$. $\alpha$ is a variable with an interval between 0 and 1, and generating a new initial noise by orthogonalizing two random Gaussian noises. Furthermore, a suite of ablation experiments is carried out to investigate the correlation between sample quality and noise percentage.

The visualization result Fig. 11 shows that as the $\alpha$ increases, the details of each target become more visually appealing. In addition, the parts of the sample Fig. 11(c) and 11(e) with red boxes indicate that the background buildings also change as the alpha changes. The background rendering ability may decline as the alpha increases, but the clarity and simulation of the image will improve, and the distribution of the initial noise is more suitable for the RS sample generation task. Combining the above, the quality of sample generation achieves the optimal state when the noise ratio reaches 1.0.

## V. CONCLUSION

We propose an efficient and controllable RSFSG based on the diffusion model. To improve the performance of the diffusion model under lightweight parameters, we design a multifrequency adaptive alignment distillation method to

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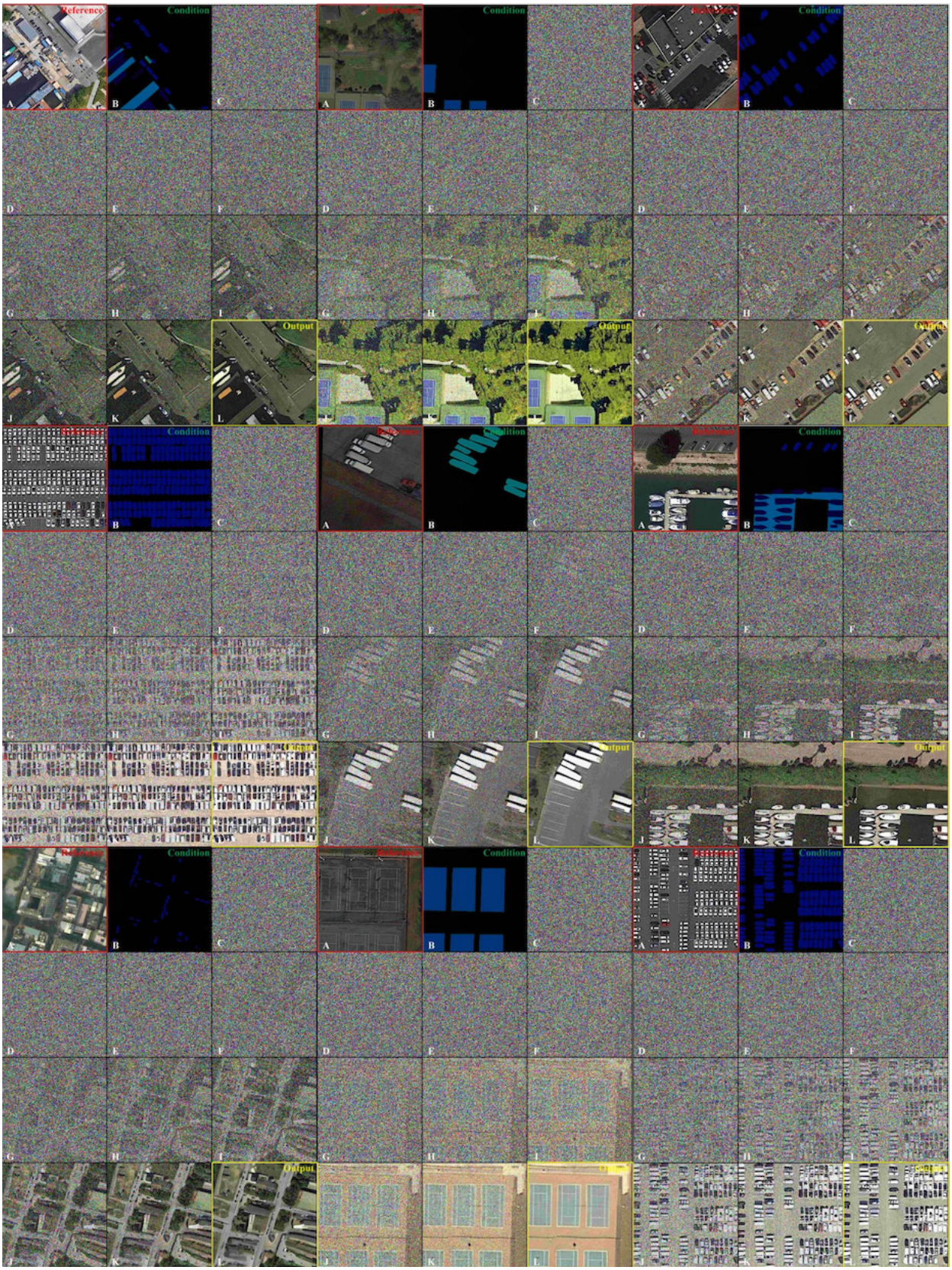



Fig. 12. Visualization of more results with the proposed efficient and controllable RSFSG framework. (a) is the GT sample, (b) is the mask, (c)-(k) are the intermediate results, (l) are the generated image.

improve the quality of fake sample generation by improving the spectral uniformity of the noise estimate. In order to reduce the training pressure of the diffusion model, we propose a progressive training method, which gradually enables the model to acquire multiscale modeling capabilities from coarse to fine. Extensive evaluations show that the proposed model

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performs favorably against state-of-the-art methods on the controllable RSFSG.

In future work, we will explore further control over the scene of the generated image, leading to superior rendering of fake objects in different situations. In addition, using semantic information such as caption to edit generated samples is also one of our directions.

## APPENDIX

See Fig. 12.

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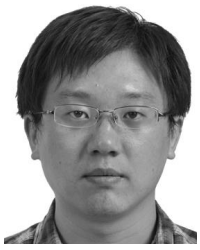
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